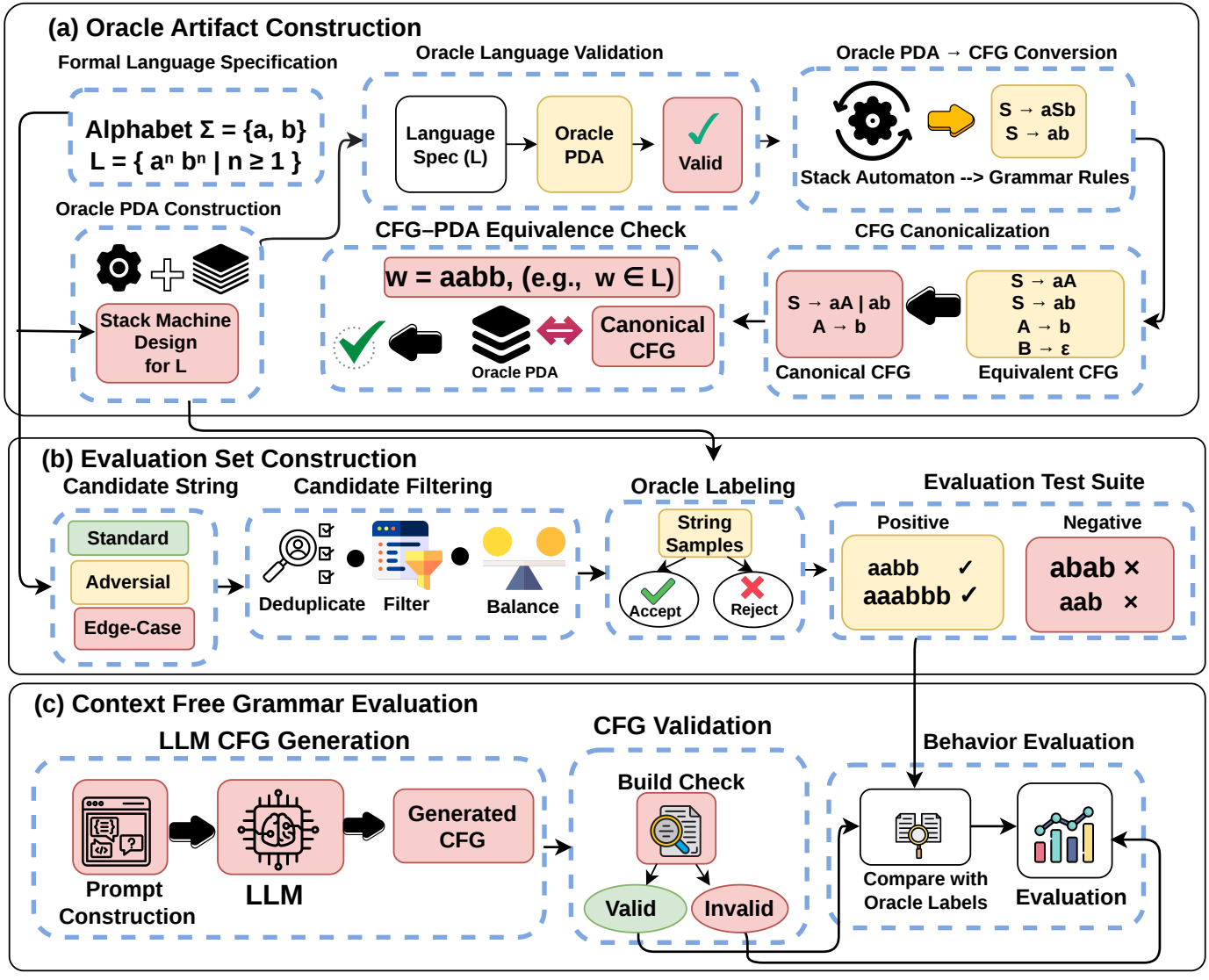
(a) Oracle Artifact Construction
Formal Language Specification
Alphabet Σ = {a, b}
L = { aⁿ bⁿ | n ≥ 1 }
Oracle Language Validation
Language Spec (L)
Oracle PDA
Valid
Oracle PDA → CFG Conversion
S → aSb
S → ab
Stack Automaton --> Grammar Rules
Oracle PDA Construction
Stack Machine Design for L
CFG–PDA Equivalence Check
w = aabb, (e.g., w ∈ L)
Oracle PDA
Canonical CFG
CFG Canonicalization
S → aA | ab
A → b
Canonical CFG
S → aA
S → ab
A → b
B → ε
Equivalent CFG
(b) Evaluation Set Construction
Candidate String
Standard
Adversial
Edge-Case
Candidate Filtering
Deduplicate
Filter
Balance
Oracle Labeling
String Samples
Accept
Reject
Evaluation Test Suite
Positive
aabb ✓
aaabbb ✓
Negative
abab ×
aab ×
(c) Context Free Grammar Evaluation
LLM CFG Generation
Prompt Construction
LLM
Generated CFG
CFG Validation
Build Check
Valid
Invalid
Behavior Evaluation
Compare with Oracle Labels
Evaluation